Body mass of Palmer penguins by species and sex

A worked example of the analysis project template

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1 Purpose

This report shows how the parts of the project template fit together on a small public dataset. Three scripts prepare the data, compare body mass between species and sexes, and gather every number this report and the slides use into one file. The report does no analysis of its own: every number below is read from that file.

2 Data

The data are measurements of adult penguins of three species near Palmer Station, Antarctica, collected from 2007 to 2009 (Gorman et al. 2014) and distributed with R's datasets package (Horst et al. 2022; Kaye et al. 2025). Of the 344 penguins, a total of 333 had every measurement and a recorded sex. The other 11 (6 Adelie and 5 Gentoo) were left out.

3 Methods

Mean body mass is given for each species and sex with a confidence interval of 95% based on the t distribution. The difference between males and females within each species comes with a Welch interval. One linear model, body mass on species and sex, estimates the difference between the sexes adjusted for species. It assumes the same difference in every species; the within-species intervals show how far that holds.

4 Results

Males were heavier than females in all three species (Figure 1, Table 1). The difference was smallest in Chinstrap penguins, at 412 g (95% CI 254 to 570), and largest in Gentoo penguins, at 805 g (95% CI 697 to 913). Adjusted for species, males were heavier by 668 g (95% CI 599 to 736), and the model explained 85% of the variance in body mass. Gentoo penguins were heavier than Adelie penguins of the same sex by 1378 g (95% CI 1301 to 1455); Chinstrap and Adelie penguins differed by 27 g (95% CI −65 to 118), an interval that includes zero.

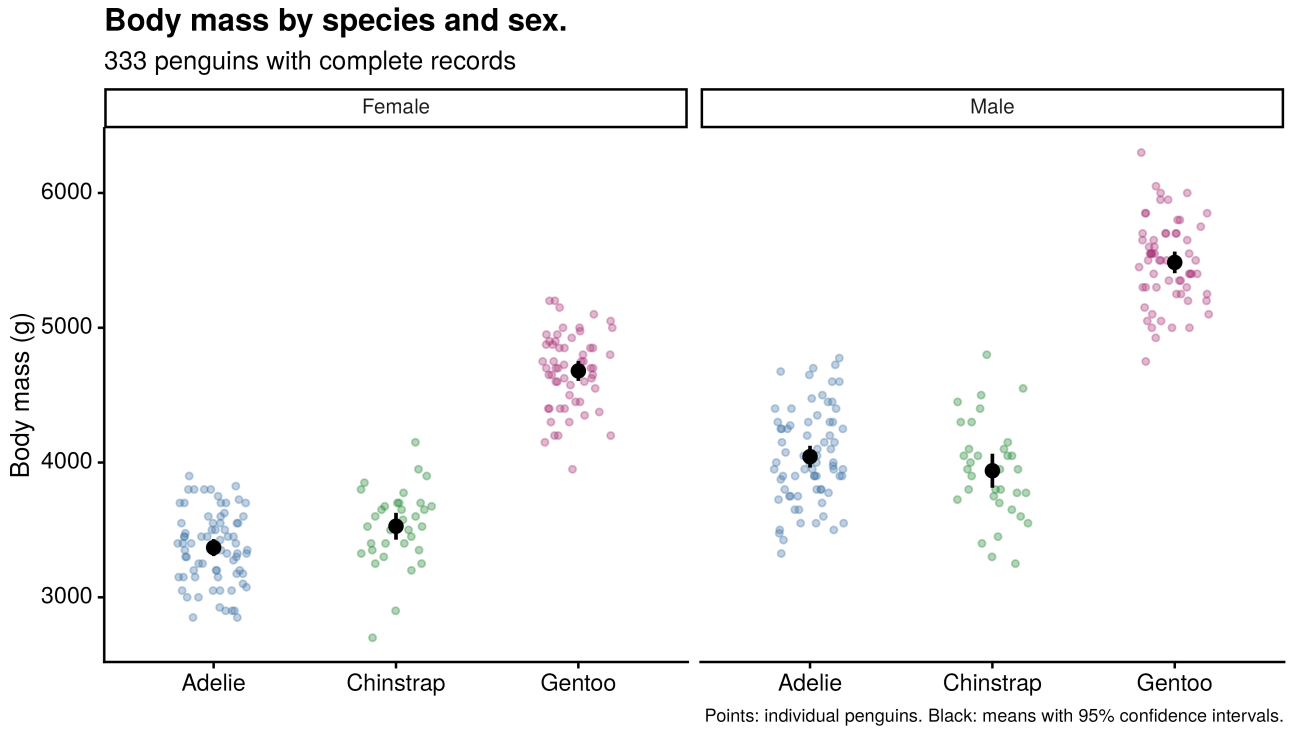


Figure 1: **Males are heavier in every species.** Body mass of each penguin, with the mean and 95% confidence interval for each species and sex (333 penguins).

Table 1: **Body mass by species and sex.** Penguins with every measurement and a recorded sex.

	Female		Male		
Species	n	Mean (95% CI)	n	Mean (95% CI)	Male minus female (95% CI)
Adelie	73	3369 (3306 to 3432)	73	4043 (3963 to 4124)	675 (573 to 776)
Chinstrap	34	3527 (3428 to 3627)	34	3939 (3813 to 4065)	412 (254 to 570)
Gentoo	58	4680 (4606 to 4754)	61	5485 (5405 to 5565)	805 (697 to 913)

Body mass in grams.

Because the within-species differences range from 412 to 805 g, the adjusted difference is an average over species rather than a value that holds in each of them.

References

Gorman, Kristen B., Tony D. Williams, and William R. Fraser. 2014. "Ecological Sexual Dimorphism and Environmental Variability Within a Community of Antarctic Penguins (Genus Pygoscelis)." *PLoS One* 9 (3): e90081. https://doi.org/10.1371/journal.pone.0090081.

Horst, Allison M., Alison Presmanes Hill, and Kristen B. Gorman. 2022. "Palmer Archipelago Penguins Data in the palmerpenguins R Package: An Alternative to Anderson's Irises." *R J.* 14 (1): 244–54. https://doi.org/10.32614/RJ-2022-020.

Kaye, Ella, Heather Turner, Kristen B. Gorman, Allison Horst, and Alison Hill. 2025. *Preparing the Palmer Penguins Data for the datasets Package in R*. Zenodo. https://doi.org/10.5281/zenodo.14902740.